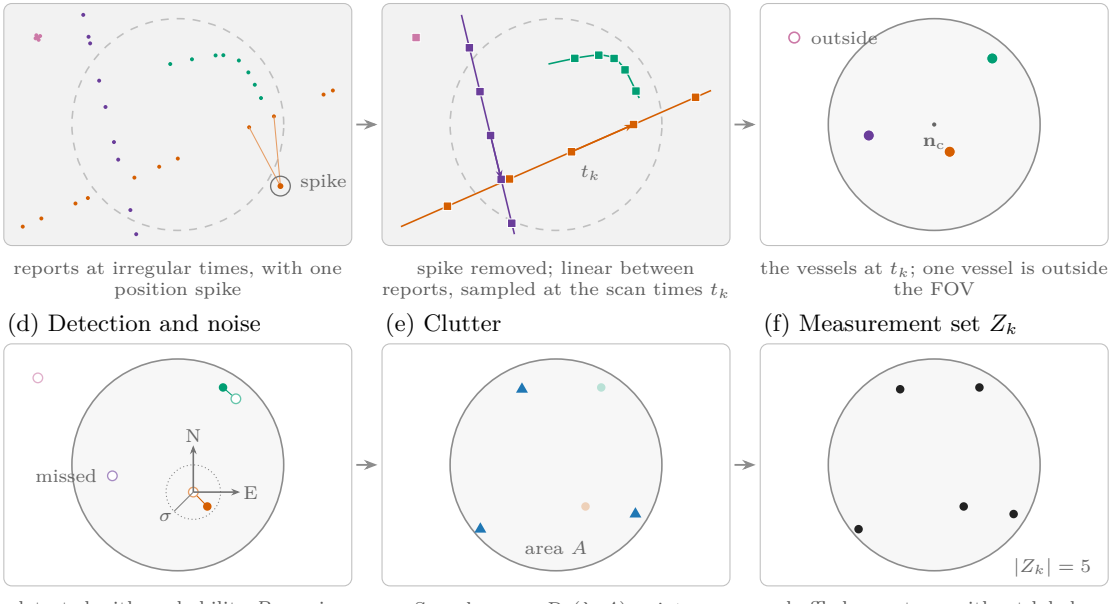

(a) AIS reports
spike
(b) Truth at the scan times
t_k
(c) Field of view at t_k
outside
$\mathbf{n}_c$
reports at irregular times, with one position spike
spike removed; linear between reports, sampled at the scan times t_k
the vessels at t_k; one vessel is outside the FOV
(d) Detection and noise
missed
N
E
σ
(e) Clutter
area A
(f) Measurement set Z_k
$|Z_k| = 5$
detected with probability P_D; noise σ in the local North-East frame
Sample $m_k \sim \mathrm{Po}(\lambda_c A)$ points, uniform on the spherical cap
shuffled n-vectors, without labels: what the tracker receives